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Discover Water

RESEARCH Open Access

Assessment of trifluoroacetic acid in tap water from Besançon (France) and bottled water from France, Italy, and Romania

Grégorio Crini[1*], Dario Lacalamita[2], Corina Bradu[3] and Chiara Mongiovi[4]

*Correspondence:
Grégorio Crini
gregorio.crini@umlp.fr
[1]Laboratoire Chrono-Environnement, Université Marie et Louis Pasteur, 16 Route de Gray, 25000 Besançon, France
[2]Dipartimento di Chimica, Università degli Studi di Bari Aldo Moro, Bari, Italy
[3]Department of Systems Ecology and Sustainability, PROTMED Research Centre, University of Bucharest, Bucharest, Romania
[4]Istituto per i Processi Chimico-Fisici, CNR, Via Orabona 4, Bari, Italy

Abstract
Trifluoroacetic acid (TFA), often referred to as the simplest "Forever Chemical," is a highly persistent environmental contaminant. Despite increasing concern about its potential toxicity, analytical data on its occurrence in drinking water remain limited. In this study, TFA was quantified in 63 household tap water samples from Besançon (France) and 71 bottled water samples purchased in France, Italy, and Romania. The 26 participating French households were recruited on a voluntary basis, and samples were collected randomly over a six-month period. TFA was detected in 100% of the tap water samples, with concentrations ranging from 540 to 3800 ng/L (mean: 1164 ± 594 ng/L; median: 1000 ng/L). In bottled water, TFA was also detected in all French samples, though at lower levels (mean: 297 ± 268 ng/L; median: 280 ng/L). Concentrations in Italian and Romanian bottled water ranged from below the detection limit to a maximum of 400 ng/L. These findings highlight the widespread presence of TFA in drinking water, with higher concentrations in tap water compared to bottled water, except for one outlier brand. The results support the need for further monitoring and assessment of TFA as an emerging contaminant in European water systems.

Keywords Trifluoroacetic acid, Drinking water, PFAS, Analytical monitoring, Regulatory framework

1 Introduction

Environmental pollution has evolved significantly over recent decades, shifting from well-known conventional contaminants to more complex and recently recognized emerging threats. Conventional contaminants such as metals, metalloids, nitrates, sulphate, phosphates, fluorides, and some organics (*e.g.*, pesticides, volatile organic compounds, polycyclic aromatic hydrocarbon, polychlorinated biphenyls) have been extensively studied and regulated, particularly on their presence in water resources. Growing concern now focuses on the so-called "emerging pollutants" including antibiotics, hormones, illicit drugs, endocrine disruptors, cosmetics, personal care products, metabolites of pesticides, surfactants, industrial products, microplastics, nanoparticles,

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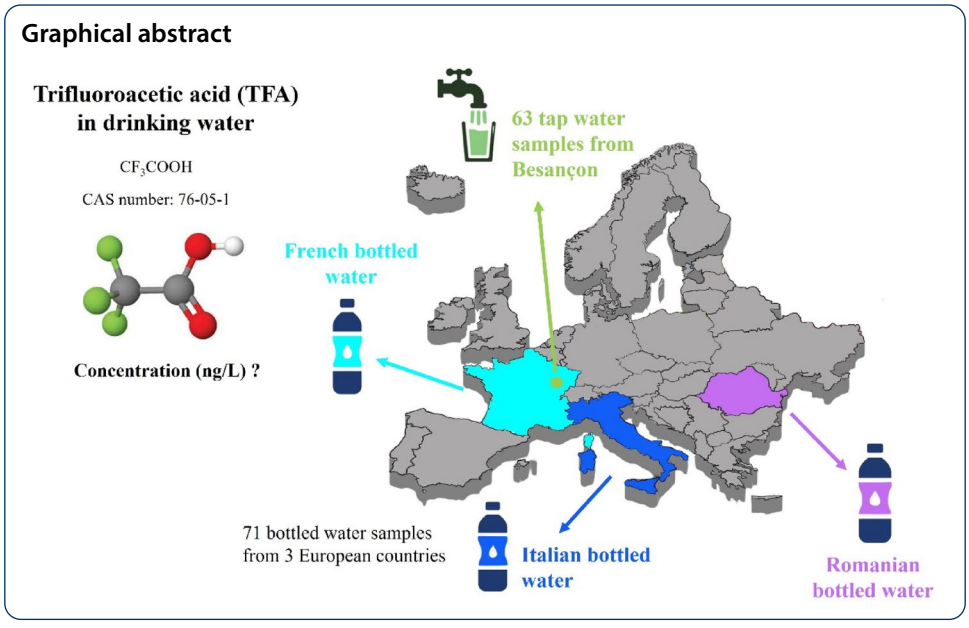

nanomaterials, radionuclides, and particularly per- and polyfluoroalkyl substances (PFAS) [1–10]. The appearance of these compounds in aqueous environmental compartments worldwide has indeed become a major concern for society and public health authorities, since these molecules may originate adverse effects on animals and humans by, for example, interfering with the normal action of the endocrine systems.

PFAS, commonly referred to as "forever pollutants" due to their persistence in the environment and high resistance to degradation, are chemical substances whose specific properties, *e.g.*, non-stickiness, waterproofness, and heat-resistance, make them widely used in many everyday products such as non-stick cookware, takeout containers, surfactants, cosmetics and personal care products, waterproof apparel, furniture, and textiles. As a result of their intensive use in industry and agriculture since the 1950s [11], all environmental compartments are contaminated by these substances, which can be transported over long distances through water bodies and atmosphere. PFAS and their metabolites have been also detected in human blood, urine, and breast milk [12]. The two main sources of human exposure to PFAS are the consumption of contaminated beverages and the ingestion of contaminated food, either from soils or through migration of PFAS from food packaging or cooking utensils. Other sources include the use of various everyday products and objects, contaminated air, dust and soil, and the occupational environment involved in the manufacture and use of these substances [13, 14]. The presence of PFAS in drinking water is a source of concern for an increasing number of people around the world, which affects the choice between drinking bottled and tap water [15–17].

Among per- and polyfluoroalkyl substances, trifluoracetic acid (TFA) has attracted special attention because it worries the scientific community as it is omnipresent in the environment, *e.g.*, rivers, groundwater, soils and plants, very persistent, and potentially toxic. Nevertheless, in Europe, this molecule is under surveillance as an emerging drinking water contaminant, whose chemical, biological and toxicological information must be consolidated [18]. TFA is an ultra-short chain PFAS containing two carbons and a

carboxylic acid function, highly mobile in the environment due to its simple chemical structure and polar moiety that give it a strong hydrophilic character (Fig. 1), and very persistent due to its high chemical stability and non-biodegradable character [19–22]. Thus, due to its stability and hydrophilicity, TFA can easily transported on long distances, reaching all environmental compartments. For example, this PFAS can easily reach drinking water sources which can then become an important human exposure route [13, 23–31]. Studies on the potential toxicity of TFA are increasingly being published, with growing evidence linking this contaminant to several cancers, elevated cholesterol levels, impaired fertility and fetal development, and adverse effects on kidney and liver function [18, 22, 29, 31–33]. However, toxicity and ecotoxicity data are still scarce and require further elaboration [18, 34].

To protect water quality, PFAS have been the subject of specific regulations in recent years that are constantly evolving based on published results [35]. In Europe, a 2020 directive imposes a quality limit set at 100 ng/L for the sum of 20 PFAS, considered as substances of concern and priority, in water intended for human consumption (European Drinking Water Directive of the European Parliament and Council of December 16, 2020, EU Directive 2021/2184). This quality limit is not a health value, but a regulatory criterion that must be met. This directive, which will come into force in January 2026 in all EU member states, also sets a limit value for "total PFAS" at 500 ng/L. The list of 20 PFAS include three compounds banned from production, use, and marketing: perfluorooctane sulfonic acid (PFOS, banned in 2009), perfluorooctanoic acid (PFOA, banned in 2020) and perfluorohexane sulfonic acid (PFHxS, banned in 2022). Surprisingly, TFA, the simplest and most mobile of the PFAS, is not included in this list, and even escapes worldwide regulation, although TFA is suspected of having harmful effects

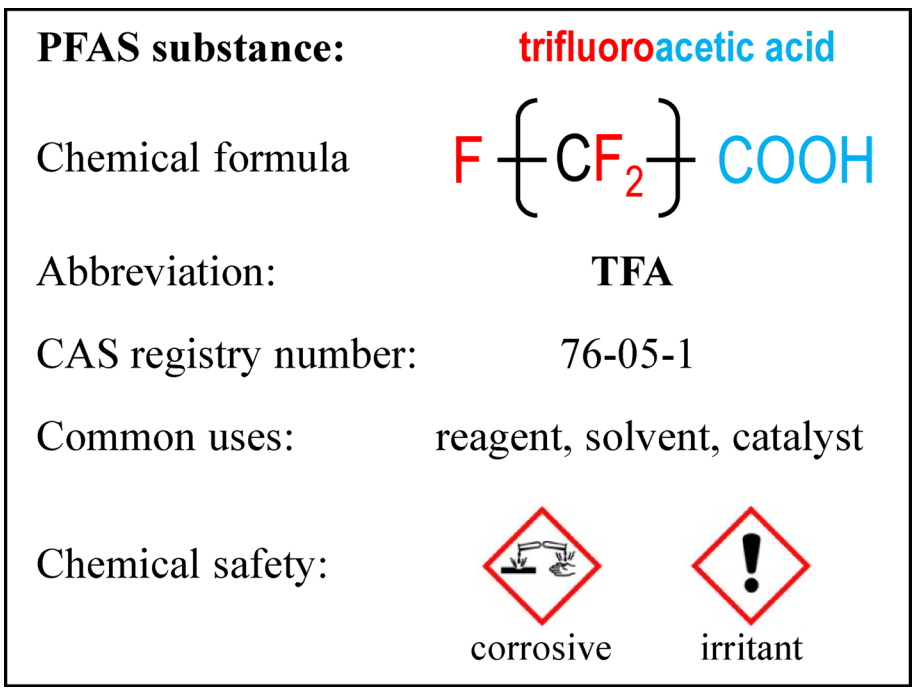

Fig. 1 Compound name, acronym, chemical formula, common uses, and chemical safety of trifluoroacetic acid, the simplest perfluoroorganic acid

on human health and is found everywhere. Despite TFA is considered a PFAS, it is not among the 20 PFAS regulated by the European Drinking Water Directive [21, 22].

The European directive has been transposed into French law and came into force on January 1, 2023. However, its application will only be effective from January 1, 2026, the date from which water distributors and regional health agencies will have to register and search for the 20 substances as part of the analyses carried out on drinking water. In a report of 2019, the French Agency for Food, Environmental and Occupational Health & Safety (ANSES) proposed a value of 900 ng/L in drinking water as an action threshold for metabolites classified as irrelevant, as is the case for some pesticide metabolite [36]. Following a recent report by the same Agency, France has classified TFA as an irrelevant metabolite [37] while Germany has classified TFA as a reprotoxic substance. However, according to ANSES, TFA should be considered "de facto" a relevant metabolite for drinking water if its concentration is higher than 100 ng/L. In this case, the water should be considered "non-compliant" with the quality standard. Thus, quality limits in drinking water should apply to TFA: the water must meet both the 500 ng/L threshold for total PFAS (including TFA), and the 100 ng/L threshold for TFA alone. All this shows that there is still no consensus on this PFAS, and efforts must therefore be mobilized to acquire chemical and biological data [18, 22, 34–38].

Within this framework, the presence of TFA, a short-chain PFAS, in drinking water has been monitored as part of the Franco-Italian-Romanian project (WATEREX 2025–2029). This project presents a novel investigation into the presence of TFA in both tap and bottled water across multiple European contexts with two main objectives: (i) to establish a comprehensive analytical map of PFAS contamination, including TFA, in tap water from Besançon, Bari, and Bucharest, in bottled water marketed in the three countries, and in effluents from selected European industries (surface treatment, paper mills, laundries) and urban wastewater treatment plants; and (ii) to propose innovative removal strategies, including biosorption, ion-exchange, and advanced oxidation through selective catalytic defluorination. In this study, TFA concentrations were measured in 63 tap water samples from households in Besançon and in 71 bottled water samples from France, Italy, and Romania. Analyses were carried out by a French accredited laboratory, with a quantification limit of 50 ng/L for clean water. To our knowledge, this is the first study to report TFA levels in household tap water in Besançon (France), as well as among the few that systematically assess TFA in bottled water consumed in France, Italy, and Romania.

2 Materials and methods

All the methods were carried out in accordance with relevant guidelines and regulations. All experimental protocols were certified by a French accredited laboratory (IANESCO, Poitiers, France; COFRAC accreditation no. 1–6209, ISO/DIS 18127).

2.1 Study area

The analysis focused on three European cities: Besançon (France; 47.2378° N, 6.0241° E), Bari (Italy; 41.1171° N, 16.8719° E), and Bucharest (Romania; 44.4268° N, 26.1025° E). This study involved only non-invasive sampling of household tap water, without collecting any personal or health-related data. In accordance with national and institutional guidelines, no formal ethical approval was required. Nevertheless, informed consent

was obtained from all participating households prior to sample collection, following relevant regulations and good research practices. In Besançon, all 63 tap water samples were collected from 26 volunteer households across six distinct districts. The selection of households was based on voluntary participation. The three cities for bottled water sampling were selected to represent diverse geographical, hydrological, and infrastructural contexts across Western, Southern, and Eastern Europe. Bottled water samples were selected from multiple commercial brands, and when possible, bottles from different production batches were included to account for intra-brand variability.

2.2 Sampling protocol

The sampling protocol followed the principles outlined in ISO 5667–5:2006 for municipal drinking water sampling [39], and best practices for PFAS sampling were also considered, including container selection and cold chain preservation during transport [40]. Tap water samples were collected from 26 volunteer households in Besançon, France, comprising an equal number of houses and apartments (n = 13 each). Sampling was conducted over a six-month period, from November 2024 to April 2025, yielding a total of 63 samples. Only one household reported the use of a domestic water filtration and softening system. Household drinking water habits varied: 17 consumed exclusively tap water, 8 used both tap and bottled water (from four different commercial brands), and 1 relied solely on bottled water. All tap water samples were collected from the kitchen cold-water tap, using the first morning draw, between 06:30 and 07:30. The procedure was standardized across all households and executed by an accredited service provider. Water samples were collected directly into sterile containers supplied by the analytical laboratory, immediately placed in a cooler maintained at 5 °C, and shipped within 24 h. Bottled water samples were obtained from retail outlets in France (n = 48, eleven brands), Italy (n = 15, three brands), and Romania (n = 8, two brands). Each sample originated from a different production batch, selected through random sampling. Only still, non-carbonated waters were included. Water was transferred aseptically from the original bottles into laboratory-provided containers, stored at 5 °C, and transported under temperature-controlled conditions.

2.3 Sample preparation, analytical analysis, and chromatographic conditions

TFA analyses were performed by an accredited laboratory (IANESCO, Poitiers, France) following a COFRAC-accredited method (COFRAC accreditation no. 1–6209, ISO/DIS 18127) [41], ensuring validated procedures for PFAS quantification in clean water using LC–MS/MS with isotope dilution and solid phase extraction. The protocol complies with the requirements of ISO 18127 for the determination of adsorbable organically bound fluorine in water, and the laboratory operates according to the quality management standards of ISO/IEC 17025 [42, 43]. Sample preparation included acidification of each sample to pH 3, addition of isotope-labelled internal standards, and extraction by solid phase extraction on an anion exchange column. After column conditioning, 250 mL of sample was loaded, rinsed with ultrapure water, and the analytes eluted with an alkaline organic solvent. The extract was concentrated under vacuum to 125 µL, then mixed with methanol prior to LC–MS/MS analysis. The analysis was carried out using an LC20-40 chromatograph coupled with an 8060 triple quadrupole mass spectrometer (Shimadzu). Compounds were separated on an XTERRA C18 chromatographic column

(Waters—50 × 3.0 mm, 3.5 µm particle size) with a methanol–water gradient (flow rate 0.5 mL/min). Injection volume was 5 µL. Detection was performed by electrospray ionization in negative mode, monitoring two specific transitions per compound (one for quantification and one for confirmation). Quantification was based on calibration curves using the response ratio between analytes and their corresponding isotope-labelled internal standards.

2.4 Quality assurance and quality control

Quality assurance and quality control procedures were ensured by the accredited laboratory (COFRAC) performing the analyses, including the use of laboratory blanks, field blanks, and recovery validation according to ISO 5667–3:2018 standards [39]. The analytical method for the determination of TFA has been rigorously validated to ensure its performance across its working range (Table 1). The validation was performed at three key concentrations: 0.05 µg/L, 0.25 µg/L, and 2.0 µg/L. The method demonstrated excellent trueness, with an overall mean recovery of 97.8% and low calculated biases at each level: 2.53% at LOQ, – 1.84% at Level 2, and – 0.69% at Level 3 (Table 1). The intermediate precision, expressed as the coefficient of variation (CV), was confirmed to be within acceptable limits, ranging from 22.62% at the LOQ to 15.66% at the highest concentration. The accuracy profile confirmed that the method was accurate at all tested levels, as all results fall within the pre-defined tolerance intervals. The expanded measurement uncertainty (U), with a coverage factor of k = 2, has been established as follows: 53% at the LOQ, 48% for concentrations between > 0.05 µg/L and 0.25 µg/L, and 43% for concentrations above 0.25 µg/L.

Ongoing quality control procedures were strictly followed to ensure the continued reliability of results.

- Calibration blanks: Analyzed in each run and was consistently below 10% of the signal of the lowest calibration standard.
- Laboratory blanks: A blank was processed with every sample batch and must be below the LOQ. These are typically observed to be less than 15 ng/L.
- LOQ verification: The limit of quantitation is verified daily at the end of each analytical sequence. If criteria are not met, the instrument undergoes maintenance, and the batch is re-analyzed.

Table 1 Summary of method validation for TFA analysis across three concentration levels for limit of quantification (LOQ): Level 1 = 0.05 µg/L, Level 2 = 0.25 µg/L, Level 3 = 2 µg/L. Validation confirms analytical reliability across tested matrices

Parameter	Level 1 (LOQ)	Level 2 (LOQ)	Level 3 (LOQ)
Concentration level tested	0.05 µg/L	0.25 µg/L	2 µg/L
Number of values	12	21	21
Maximum acceptable error	60%	40%	40%
Trueness			
Mean recovery	100.25%	95.98%	97.11%
Calculated bias	2.53%	– 1.84%	– 0.69%
Precision			
Intermediate precision of the coefficient of variation	22.62%	16.95%	15.66%
Overall accuracy			
Method meets accuracy criteria?	Yes	Yes	Yes
Measurement uncertainty			
Expanded uncertainty (coverage factor = 2)	53%	42%	44%

Table 2 Summary of the main statistical values of trifluoroacetic acid in French tap water samples (63 samples; detection frequencies = 100%)

Minimum (ng/L)	Maximum (ng/L)	Mean (ng/L)	Median (ng/L)
540	3800	1164 ± 594	1000

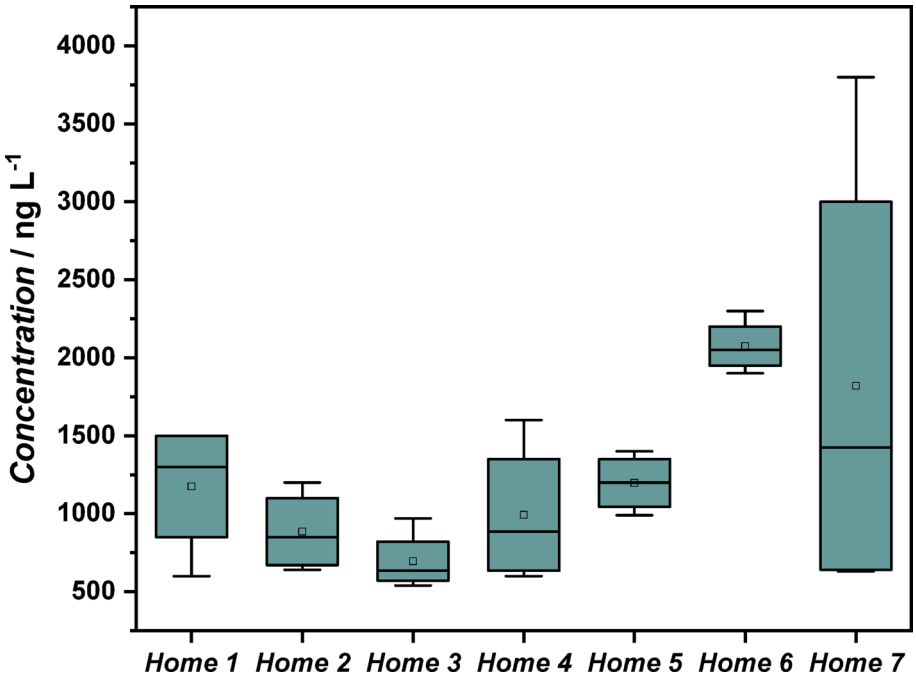

Fig. 2 Concentration of trifluoroacetic acid tap water samples collected from seven French households that participated in the study (Boxes represent the interquartile range (IQR, 25th to 75th percentile), the square symbol indicates the mean, the horizontal line within the box is the median, and whiskers extend to the minimum and maximum values within 1.5 times the IQR)

3 Results and discussion

3.1 Trifluoroacetic acid in tap water samples from Besançon

The measured concentrations of TFA in tap water samples from Besançon reveal important insights into the current exposure levels of this emerging contaminant in urban drinking water. TFA was detected in 100% of the 63 tap water samples collected from 26 French households, with concentrations ranging from 540 to 3,800 ng/L (Table 2). The average concentration was 1,164 ± 594 ng/L, with a median value of 1,000 ng/L. Considerable variability was observed among households, as reflected by the relatively high standard deviation.

Figure 2 illustrates the measured concentrations of trifluoroacetic acid in tap water samples collected from seven different French households. Specifically, Home 7 shows the widest distribution and the highest maximum concentration, approaching 3800 ng/L, suggesting a potentially significant or intermittent source impacting that specific tap water supply. Conversely, households like Home 3 exhibit consistently lower TFA levels, with a narrower range centred around 600–700 ng/L. The presence of outliers or highly skewed distributions, particularly evident in home 7, could indicate specific local contamination events or unique water treatment/distribution dynamics.

Further emphasizing these differences, Table 3 presents the repeatability of measurements conducted in these seven households under identical conditions and within a

Table 3 Repeatability of the measurements taken on tap water from seven households (samples, sampling protocols and analysis conditions identical; concentrations expressed in ng/L)

	Home 1	Home 2	Home 3	Home 4	Home 5	Home 6	Home 7
Sample 1	1500	1200	600	1600	1400	2000	650
Sample 2	600	1000	670	1100	1300	2100	630
Sample 3	1100	640	540	670	1100	1900	3800
Sample 4	1500	700	970	600	990	2300	2200
Average	1175 ± 427	885 ± 262	695 ± 191	993 ± 461	1198 ± 186	2075 ± 171	1820 ± 1511

4-month interval. The data consistently highlight the variation in tap water quality from one household to another. For instance, the average TFA concentrations for Home 3 and Home 6 are 695 ± 191 ng/L and 2075 ± 171 ng/L, respectively, confirming the significant disparities observed across the study sites.

The average and median values observed in this study are comparable to those reported by Dong et al. [44] who analysed 9 tap waters from Chinese households in eight cities of China (mean concentration of 1712 ± 174 ng/L; detection frequency of 100%). Other studies also reported similar levels and detection frequency of TFA in drinking water. Jiao et al. [45], studying the presence of TFA in 39 tap water samples from Shanghai, China, also reported that TFA was present in 91% in samples, with concentrations ranging from 1740 to 7080 ng/L. Zheng et al. [12] stated that TFA was the dominant PFAS in US drinking water (median of 79 ng/L) with a detection of frequency of 95%. Sadia et al. [34] reported that ultrashort chain (C2-C3) PFAS, mainly TFA, are by far the most dominant class found in both Dutch raw and produced drinking water samples, with concentrations ranging from 3 to 1,100 ng/L. Arp et al. [18] indicated that TFA concentrations in the environment are orders of magnitude higher than those of other PFAS.

Other studies have shown much higher values of TFA in tap water due to the presence of industrial activities releasing into the environment where the water is taken. For example, Neuwald et al. [46] indicated that TFA was the predominant PFAS in 46 German water samples for more than 90% of the total concentration of PFAS, with a maximum and median concentration of 12,400 and 900 ng/L, respectively, in agreement with the results previously published by Scheurer et al. [20].

All these findings as well as the data in this study show that TFA is very present in tap water but that the concentration can vary significantly, depending on the local context. Several hypotheses can indeed be made. Nevertheless, no official data are available regarding the detection of PFAS in the spring water of the city of Besançon., which is served by three different water distribution networks, with five water sampling sources including four underground. However, the TFA will not be part of this list. The hypothesis of contamination at the source is not considered in this case. In addition, only one household uses a tap water filtration and softening system. The differences can then be explained by the type and state of deterioration of the pipes of the water distribution network, and that of buildings and houses (for example, those using plastic). However, it is difficult to establish a possible correlation without further measurements and information.

The literature offers different interpretations to explain both the presence and high concentrations of TFA. Zhai et al. [47] reported an increase of TFA levels from not detected in 2002 to 155 ng/L in 2012 in tap water in China in relation to the increase in anthropogenic activities, mainly industry. Li et al. [25] also reported that, in China,

PFAS pollution levels in drinking water were positively influenced by the existence of industrial sources close to water resources. Endirlik et al. [24], studying the assessment of PFAS levels in tap and bottled water samples from Turkey, showed that contamination of tap water was related to contamination in its source.

Neuwald et al. [46] indicated that TFA can be introduced into the water cycle not only through industrial processes and a transformation product of agricultural and pharmaceutical substances, but also as a transformation product of hydrofluorocarbon refrigerants in the atmosphere, which can then reach the aqueous environment via atmospheric deposition.

More recently, high concentrations of TFA were positively correlated with the degree of urbanization/industrialization in the catchment area as reported by Zhi et al. [48]. The median concentration of TFA in drinking water (160 ng/L) is about 2 orders of magnitude higher than that of perfluoropropanoic acid (2.5 ng/L). TFA was detected at a high level (11,000 ng/L) in a German tap water sample and industrial discharges into the river have been identified as the source of the contamination [48].

3.2 Trifluoroacetic acid in drinking-water consumption: tap versus bottled water

In plastic bottled water samples, TFA was also present in 100% of French samples analysed (Table 4), but at lower concentrations than in tap water. The average value was 297 ± 268 ng/L (median = 280 ng/L), four times lower than the average of tap water samples, *i.e.*, 1164 ± 594 ng/L.

Among the 48 water bottles analysed, the highest value was 1000 ng/L, which is 4 times lower than the maximum value found in a sample of tap water (3800 ng/L). Similarly, the minimum value found for bottled water (50 ng/L) is 10 times smaller than that of tap water (540 ng/L). However, this high maximum (1000 ng/L) was only found for one brand of bottled water. When the highest concentrations are excluded, the resulting average is 229 ± 204 ng/L and the median of 130 ng/L.

Table 5 shows the data on the repeatability of the measurements made on the bottled water of several commercial brands. Analyses were also conducted on French commercial water samples packaged in glass bottles. None of the 8 samples contained TFA (value less than 50 ng/L).

The interpretation of data showed, that for all samples analysed, TFA levels in plastic bottled water samples were lower than in tap waters. A similar conclusion was published by Dong et al. [44] in their study on the comparison of TFA levels in tap and bottled water from China. They reported a TFA average concentration of 6.6 ± 9.6 ng/L for six Chinese bottled from six different manufacturers with a detection frequency of 100%. As with tap water, TFA is certainly identified in bottled water but the concentrations can also vary significantly from batch to batch.

The differences could be explained by the depth of the source used to collect bottled water and by the type of plastic used to produce the bottles. In France, no

Table 4 TFA in bottled water from France, Italy, and Romania: detection frequency (%), concentration range, mean, and median (ng/L); calculations exclude values below quantification limit

Country	Number of samples	Detection frequency	Minimum	Maximum	Mean	Median
France	48	100	50	1000	297 ± 268	280
Italy	15	53	67	400	212 ± 138	200
Romania	8	0				

Table 5 Repeatability of measurements on bottled water from various commercial brands (each line = one brand; samples from different bottles; concentrations in ng/L)

Country	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
France	330	320	320	340	
France	819	800	610	710	650
France	280	280	290		
France	430	470	490		
France	820	1000	750	650	710
France	150	210	50	50	
Italy	< 50	< 50	< 50	< 50	< 50
Italy	67	86	110	84	
Italy	320	400	340	290	
Romania	< 50	< 50	< 50	< 50	< 50
Romania	< 50	< 50	< 50		

physicochemical treatment of spring water is allowed to obtain the name of natural water, excluding post-bottling contamination by the type of water treatment but hypothesizing contamination by the container. Llorca et al. [49] hypothesized that the plastic of bottles could be a source of PFAS. Schwanz et al. [50] suggested that dopants used in polymer production of water bottle manufacture and label ink labels could be responsible for water contamination. At present, no correlation could be established between the depth of the source and the TFA concentrations observed.

3.3 Trifluoroacetic acid levels in bottled water samples from France, Italy and Romania

Data obtained from bottled water samples in France were compared with those from two other European countries. TFA was analysed in 23 plastic bottled water samples collected from retail sources in Bari (Italy) and Bucharest (Romania). For the two Romanian brands and for one Italian brand out of the three analysed, no concentration of TFA could be quantified, values being below the quantification limit set at 50 ng/L (Table 5). Only two Italian brands of bottled water showed quantifiable TFA concentrations (detection frequency of 53%), ranging from undetectable to 400 ng/L, with an average of 212 ± 138 ng/L (Table 4), value comparable to that of French bottled water samples.

The heatmap (Fig. 3) provides a clear visual comparison of the TFA concentration across different bottled water brands and countries of origin. Samples from France, especially France 2 and France 5, show markedly elevated concentrations, while those from Romania consistently display values below the limit of detection (50 ng/L). Italian samples exhibit moderate variability, with Italy 3 showing the highest values among the Italian group. Overall, the heatmap emphasizes both geographic variability and potential brand-specific trends in contaminant or mineral content. Such variability may be influenced by the natural characteristics of the water sources or differences in bottling and treatment processes. These insights are especially relevant for regulatory monitoring and consumer awareness, particularly where vulnerable populations or specific dietary concerns are involved.

The absence of TFA could be correlated with the depth of the source, as one of the two Romanian sources is deep (close to 100 m), in a protected environment, and therefore less exposed to anthropogenic contamination. Similarly, for French bottled waters, it would be interesting to link the depth of the source with the values found. For the moment, with the data of this work, we have not been able to establish a correlation between the depth of the French source and the TFA values found, although certain

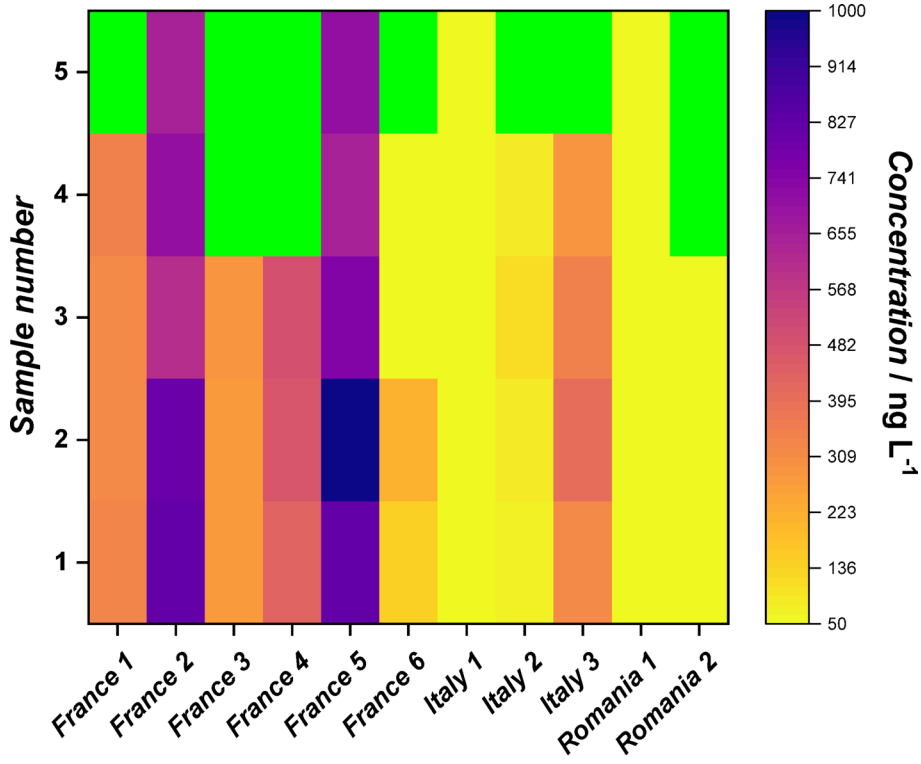

Fig. 3 Heatmap of bottled water sample values collected from different geographic origins. Each column represents a specific product group (e.g., France 1, Italy 2), while each row corresponds to an individual sample (The colour gradient indicates the concentration or intensity of the measured parameter, with blue representing higher values and yellow lower ones; the lower threshold is set at the limit of detection, defined as 50 ng/L; green indicates missing data or values not available for a given sample)

marketing arguments indicate protected water production sources or waters naturally filtered by volcanic or ice layers.

3.4 Interpretation of trifluoroacetic acid fate

Several factors may contribute to the variability in TFA concentrations observed among household tap water samples in Besançon. These include differences in pipe materials (*e.g.*, PVC, copper, lead), distance from the water treatment plant, local distribution infrastructure, and possible infiltration from nearby industrial or urban sources. Household plumbing systems may also differ in age, maintenance, or material degradation, which can affect the mobilization of contaminants. Temporal factors (*e.g.*, high rainfall or drought) may also influence contaminant concentration through dilution or concentration effects. Furthermore, the role of pesticides as a source of trifluoroacetic acid contamination is recognized but not yet fully elucidated. It is known that the degradation of pesticides containing -CF3 groups can lead to TFA formation. Denmark has just banned (July 2025) six pesticides, including 3 fungicides (fluazinam fluopyram mefentrifluconazole), 2 insecticides (flonicamid, tau-fluvalinate) and one herbicide (diflufenican), containing PFAS due to the risk of groundwater contamination [52]. In the environment, these substances are converted into TFA [51]. However, the specific pesticide classes contributing to this process remain poorly characterized due to limited data. This uncertainty highlights the need for further targeted research to clarify the pathways and extent of pesticide-derived TFA in the environment. Such investigations will be crucial

to better interpret the widespread occurrence of TFA observed in both tap and bottled water samples and to inform more effective monitoring and regulatory approaches [22, 29, 31].

The absence of detectable TFA in several Romanian bottled water samples and one Italian brand may reflect different production processes, source water characteristics, or stricter filtration technologies used by those specific producers. It is also possible that the raw water used for bottling in these cases comes from more protected aquifers with limited industrial or agricultural contamination. Additionally, batch-specific factors, such as different bottling facilities or treatment steps, may influence the final TFA levels.

To account for possible variability linked to storage or packaging, it should be noted that bottled water samples were collected from commercial outlets and may originate from different production batches. Although storage conditions were not standardized prior to purchase, all bottles were stored in cool, dark conditions before analysis. Potential leaching of TFA from plastic packaging cannot be ruled out, especially for older bottles or under elevated temperatures. However, all samples were within their shelf life at the time of analysis. No data are currently available on the contribution of bottle material or storage duration to TFA concentrations, which should be addressed in future studies.

Although the results provide valuable preliminary insights, the sampling coverage remains limited. Tap water data were collected exclusively from one city (Besançon, France), while only bottled water samples were analysed in Italy and Romania. Furthermore, the household sample size, based on voluntary participation, may not fully capture the variability across the entire urban water network. Hence, there is a clear need to expand long-term monitoring not only in the city of Besançon but also across other European cities.

3.5 Regulatory framework for drinking water

Today, neither the European Union (EU) nor the World Health Organization (WHO) have set a binding limit value for TFA (except for Denmark, which has set a binding and health-based limit value for TFA in drinking water at 9,000 ng/L), hence monitoring is not systematically required. As a result, data availability remains scarce. Nevertheless, values have been proposed and are being discussed.

There are different proposals for health guide values ranging from 2,200 to 100,000 ng/L, with a 'suggested' value of 6,000 ng/L (this value was recently recommended by the WHO) while the German Federal Environment Agency proposed a drinking water health guidance value for TFA of 60,000 ng/L [48]. The French Agency for Food, Environmental and Occupational Health & Safety (ANSES) recently recommended the same value of 60,000 ng/L, with the aim in the coming years to reduce it to 10,000 ng/L [51]. For now, in France, there is no regulation on TFA.

In Europe, if TFA was classified as a relevant metabolite in the coming years under the Drinking Water Directive, European Member States would have to comply with a limit of 100 ng/L for the presence of this molecule in drinking water. This value is identical to that of the Σ_{20}PFAS (for now, TFA not being part of this list) and the threshold value for pesticide metabolites which are occasionally sought in French waters. This proposal of 100 ng/L is simply an administrative regulatory value, unrelated to the protection of public health. In fact, the current EU legislation also considers another threshold the "total of PFAS" which extends the list of 20 substances of interest. The value of this

Table 6 Concentration or concentration ranges (means expressed in ng/L) and detection frequency (in %) of trifluoroacetic acid in tap and bottled water (TW and BW, respectively) samples from various countries

Water	Source	Country	Number of samples	TFA	Detection frequency	Reference
TW		China	39	1,174–7,080	91	[45]
TW		China	9	1,712 ± 174	100	[44]
TW	Groundwater	France	63	1,164 ± 594	100	This study
TW		China		155[a]		[47]
TW		Germany	22	50–610	100	[50]
TW		Ireland		< 18,100		[53]
TW		Germany	46	900[a]	90	[46]
TW	Surface water	The Netherlands	11	351 ± 305	100	[34]
TW	Groundwater	The Netherlands	7	245 ± 167	100	[34]
TW		Canada	7	80–130	100	[54]
TW		United States	81	79[a]	74	[12]
TW		China	1	74	100	[55]
TW		United States		41-150[a]		[56]
TW	Groundwater	Switzerland	6	16–123		[57]
BW		China	6	6.6 ± 9.6	33	[44]
BW	Groundwater	Romania	8	< 50	0	This study
BW	Groundwater	France	48	297 ± 268	100	This study
BW	Groundwater	Italy	15	212 ± 138	53	This study

[a]median concentration

regulatory threshold cannot exceed 500 ng/L, leaving each state to choose whether or not to include TFA. France did not choose this threshold but that of 100 ng/L, only for the 20 regulatory substances, excluding TFA.

Based on the results obtained, if we apply the value of 60,000 ng/L recommended by ANSES Agency, or that of 9,000 ng/L applied by Denmark or 6,000 suggested by WHO, 100% of the water samples would be compliant. If we consider the quality limit suggested by the European Union of 500 ng/L for 'total PFAS', all the concentrations of TFA alone in tap water exceeds this limit, the average value for Besançon tap water being 2 times higher. Similarly, if the limit value of 100 ng/L applied for pesticides, in particular for relevant metabolites, were to apply to TFA, 100% of Besançon tap water samples would be non-compliant, while only 40% of French bottled waters would be.

There is abundant literature on the presence of PFOA and PFOS, two PFAS that have been banned, in water in general. In contrast, data on the identification and quantification of TFA are still lacking and need to be completed, this makes comparison of our results with other studies difficult. Nevertheless, a few published works have been compiled in Table 6. These data compare the concentrations or concentration range for TFA found in tap water and bottled water from several countries. The data from this study are comparable and indicate that TFA is significantly more prevalent in tap water than in bottled water.

4 Conclusions

This study provides the first published data on trifluoroacetic acid (TFA) contamination in tap and bottled water in Europe. TFA was detected in 100% of tap water samples from Besançon (France), with concentrations up to 3,800 ng/L, indicating widespread exposure and significant variability between households. All French bottled water samples also contained TFA, but at lower levels. In contrast, several Italian and Romanian

bottled waters showed no detectable TFA. These findings highlight tap water as a primary source of exposure and reveal marked geographical and product-based differences. The widespread presence of TFA in drinking water raises important public health concerns due to its persistence and potential toxicity, emphasizing the need for continuous monitoring and inclusion in regulatory frameworks. This study's data can support water utilities and policymakers in optimizing treatment processes to reduce TFA contamination. Additionally, the validated analytical methods provide a foundation for future monitoring programs and risk assessments. Future research will broaden both the spatial and analytical scope, including a wider range of PFAS compounds and bottled water samples from France, Italy, Romania, Belgium, Portugal, and China, to better assess geographic variability and potential contamination sources. A comprehensive risk assessment of TFA will also be undertaken to better understand its environmental fate and potential health implications.

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Author contributions
G.C. designed and performed the experiments, supervised the project, and oversaw composing the first draft of the manuscript, collecting of references, and editing the final draft. D.L. contributed to data collection, prepared the tables, and was in charge for creating the figures and analyzing the coding of the references. C.B. contributed to the conceptualization of the study, was responsible for choosing the pertinent references, and assessing the analysis in the publication. C.M. was responsible for creating the work plan, the graphical abstract, assisting in conceptualization, editing and reviewing the entire document. All authors edited, read and approved the final manuscript.

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Data availability
The data supporting the findings of this study are provided within the manuscript.

Declarations

Human or animal rights
Not applicable.

Informed consent
Informed consent was obtained from all subjects and/or their legal guardian(s).

Consent for publication
All methods were carried out in accordance with relevant guidelines and regulations. All experimental protocols were approved by a French accredited laboratory (IANESCO, Poitiers, France; COFRAC accreditation no. 1–6209, ISO/DIS 18127). All authors agree on the version of the paper.

Competing interests
The authors declare no competing interests.

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